

# LINUX COMMANDS CHEATSHEET

**LAST MINUTE REVISION GUIDE FOR  
INTERVIEWS**



## 1 - SYSTEM INFORMATION

|                                      |                                                    |
|--------------------------------------|----------------------------------------------------|
| <code>uname -a</code>                | # Display Linux system information                 |
| <code>uname -r</code>                | # Display kernel release information               |
| <code>cat /etc/redhat-release</code> | # Show which version of Red Hat installed          |
| <code>lsb_release -a</code>          | # Show which version of Ubuntu installed           |
| <code>uptime</code>                  | # Show how long the system has been running + load |
| <code>hostname</code>                | # Show system host name                            |
| <code>hostname -I</code>             | # Display all local IP addresses of the host       |
| <code>last reboot</code>             | # Show system reboot history                       |
| <code>date</code>                    | # Show the current date and time                   |
| <code>cal</code>                     | # Show this month's calendar                       |
| <code>w</code>                       | # Display who is online                            |
| <code>whoami</code>                  | # Who you are logged in as                         |

## 2 - HARDWARE INFORMATION

|                                    |                                                                                |
|------------------------------------|--------------------------------------------------------------------------------|
| <code>dmesg</code>                 | # Display messages in kernel ring buffer                                       |
| <code>cat /proc/cpuinfo</code>     | # Display CPU information                                                      |
| <code>cat /proc/meminfo</code>     | # Display memory information                                                   |
| <code>free -h</code>               | # Display free and used memory ( -h for human readable, -m for MB, -g for GB.) |
| <code>lspci -tv</code>             | # Display PCI devices                                                          |
| <code>lsusb -tv</code>             | # Display USB devices                                                          |
| <code>dmidecode</code>             | # Display DMI/SMBIOS (hardware info) from the BIOS                             |
| <code>hdparm -i /dev/sda</code>    | # Show info about disk sda                                                     |
| <code>hdparm -tT /dev/sda</code>   | # Perform a read speed test on disk sda                                        |
| <code>badblocks -s /dev/sda</code> | # Test for unreadable blocks on disk sda                                       |
| <code>lshw</code>                  | # Display information about CPU, memory, storage, and network interfaces       |

lsblk

# Display information about all storage devices

### 3 - PERFORMANCE MONITORING AND STATISTICS

top

# Display and manage the top processes

htop

# Interactive process viewer (top alternative)

mpstat 1

# Display processor related statistics

vmstat 1

# Display virtual memory statistics

iostat 1

# Display I/O statistics

tail -100 /var/log/messages

# Display the last 100 syslog messages (Use /var/log/syslog for Debian based systems.)

tcpdump -i eth0

# Capture and display all packets on interface eth0

tcpdump -i eth0 'port 80'

# Monitor all traffic on port 80 ( HTTP )

lsof

# List all open files on the system

lsof -u user

# List files opened by user

free -h

# Display free and used memory ( -h for human readable, -m for MB, -g for GB.)

watch df -h

# Execute "df -h", showing periodic updates

mpstat

# Display statistics about CPU usage

pidstat

# Display statistics about processes running

### 4 - USER INFORMATION AND MANAGEMENT

id

# Display the user and group ids of your current user.

last

# Display the last users who have logged onto the system.

who

# Show who is logged into the system.

w

# Show who is logged in and what they are doing.

groupadd test

# Create a group named "test".

useradd -c "John Smith" -m john

# Create an account named john, with a

`userdel john`

comment of "John Smith" and create the user's home directory.

`usermod -aG sales john`

# Delete the john account.

# Add the john account to the sales group

## 5 - FILE AND DIRECTORY COMMANDS

`ls -al`

# List all files in a long listing (detailed) format

`pwd`

# Display the present working directory

`mkdir directory`

# Create a directory

`rm file`

# Remove (delete) file

`rm -r directory`

# Remove the directory and its contents recursively

`rm -f file`

# Force removal of file without prompting for confirmation

`rm -rf directory`

# Forcefully remove directory recursively

`cp file1 file2`

# Copy file1 to file2

`cp -r source_directory destination`

# Copy source\_directory recursively to destination. If destination exists, copy source\_directory into destination, otherwise create destination with the contents of source\_directory.

`mv file1 file2`

# Rename or move file1 to file2. If file2 is an existing directory, move file1 into directory file2

`ln -s /path/to/file linkname`

# Create symbolic link to linkname

`touch file`

# Create an empty file or update the access and modification times of file.

`cat file`

# View the contents of file

`less file`

# Browse through a text file

`head file`

# Display the first 10 lines of file

`tail file`

# Display the last 10 lines of file

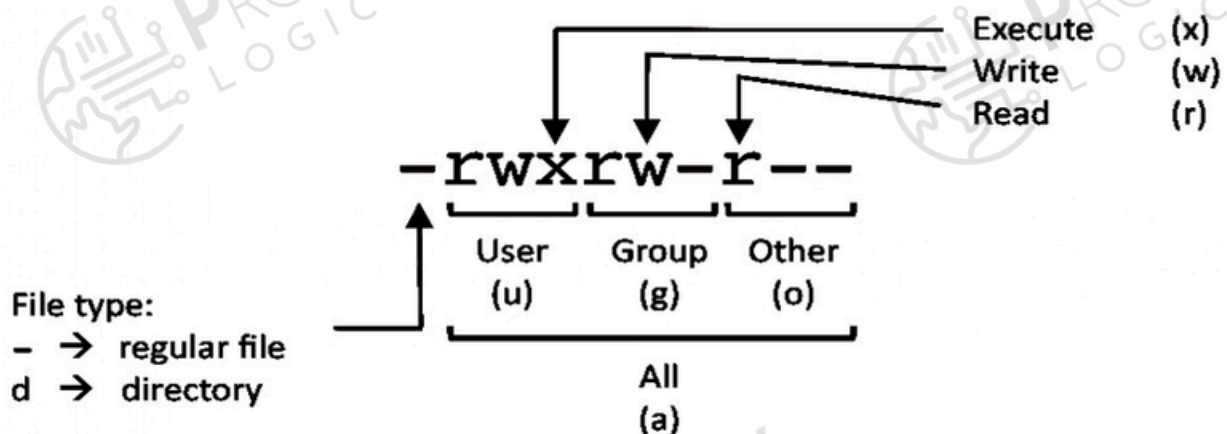
`tail -f file`

# Display the last 10 lines of file and "follow" the file as it grows.

## 6 - PROCESS MANAGEMENT

|                                        |                                                              |
|----------------------------------------|--------------------------------------------------------------|
| <code>ps</code>                        | # Display your currently running processes                   |
| <code>ps -ef</code>                    | # Display all the currently running processes on the system. |
| <code>ps -ef   grep processname</code> | # Display process information for processname                |
| <code>top</code>                       | # Display and manage the top processes                       |
| <code>htop</code>                      | # Interactive process viewer (top alternative)               |
| <code>kill pid</code>                  | # Kill process with process ID of pid                        |
| <code>killall processname</code>       | # Kill all processes named processname                       |
| <code>program &amp;</code>             | # Start program in the background                            |
| <code>bg</code>                        | # Display stopped or background jobs                         |
| <code>fg</code>                        | # Brings the most recent background job to foreground        |
| <code>fg n</code>                      | # Brings job n to the foreground                             |
| <code>nohup processname</code>         | # Runs a process even after user logs out                    |

## 7 - FILE PERMISSIONS



### PERMISSION

| U                | G                | W                |
|------------------|------------------|------------------|
| <code>rwX</code> | <code>rwX</code> | <code>rwX</code> |
| <code>rwX</code> | <code>rwX</code> | <code>r-X</code> |
| <code>rwX</code> | <code>r-X</code> | <code>r-X</code> |
| <code>rw-</code> | <code>rw-</code> | <code>r--</code> |

### EXAMPLE

```

chmod 777 filename # Use sparingly!
chmod 775 filename
chmod 755 filename
chmod 664 filename
  
```



rw- r-- r--      chmod 644 filename

#### LEGEND

U = User

G = Group

W = World

r = Read

w = write

x = execute

- = no access

chown john /path/to/file

# Change ownership of /path/to/file to john

chgrp sales /path/to/file

# Change group ownership of /path/to/file to group sales

## 8 - NETWORKING

ip a

# Display all network interfaces and IP address

ip addr show dev eth0

# Display eth0 address and details

ethtool eth0

# Query or control network driver and hardware settings

ping host

# Send ICMP echo request to host

whois domain

# Display whois information for domain

dig domain

# Display DNS information for domain

dig -x IP\_ADDRESS

# Reverse lookup of IP\_ADDRESS

host domain

# Display DNS IP address for domain

hostname -i

# Display the network address of the host name.

hostname -I

# Display all local IP addresses of the host.

wget http://domain.com/file

# Download http://domain.com/file

netstat -nutlp

# Display listening tcp and udp ports and corresponding programs

|              |                                                |
|--------------|------------------------------------------------|
| ifconfig     | # Display information about network interfaces |
| tracert host | # Display the path that packets take to host   |
| tcpdump      | # Capture and analyze network traffic          |

## 9 - ARCHIVES (TAR FILES)

|                                   |                                                          |
|-----------------------------------|----------------------------------------------------------|
| tar cf archive.tar directory      | # Create tar named archive.tar containing directory.     |
| tar xf archive.tar                | # Extract the contents from archive.tar.                 |
| tar czf archive.tar.gz directory  | # Create a gzip compressed tar file name archive.tar.gz. |
| tar xzf archive.tar.gz            | # Extract a gzip compressed tar file.                    |
| tar cjf archive.tar.bz2 directory | # Create a tar file with bzip2 compression               |
| tar xjf archive.tar.bz2           | # Extract a bzip2 compressed tar file.                   |

## 10 - INSTALLING PACKAGES

|                            |                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------|
| yum search keyword         | # Search for a package by keyword.                                                  |
| yum install package        | # Install package.                                                                  |
| yum info package           | # Display description and summary information about package for RHEL based systems. |
| rpm -i package.rpm         | # Install package from local file named package.rpm                                 |
| yum remove package         | # Remove/uninstall package for RHEL based systems.                                  |
| yum update package         | # Update package with name package for RHEL based systems.                          |
| tar zxvf sourcecode.tar.gz | # Install software from source code.                                                |
| cd sourcecode              |                                                                                     |
| ./configure                |                                                                                     |
| make                       |                                                                                     |
| make install               |                                                                                     |

|                                      |                                                                                    |
|--------------------------------------|------------------------------------------------------------------------------------|
| <code>apt-get update</code>          | # Update package list for Debian based systems.                                    |
| <code>apt-get upgrade</code>         | # Upgrade all installed packages to their newest version for Debian based systems. |
| <code>apt-get install package</code> | # Install package with name package for Debian based systems.                      |
| <code>apt-remove package</code>      | # Remove package with name package for Debian based systems.                       |

## 11 - SEARCH

|                                              |                                                                                |
|----------------------------------------------|--------------------------------------------------------------------------------|
| <code>grep pattern file</code>               | # Search for pattern in file                                                   |
| <code>grep -r pattern directory</code>       | # Search recursively for pattern in directory                                  |
| <code>locate name</code>                     | # Find files and directories by name                                           |
| <code>find /home/john -name 'prefix*'</code> | # Find files in /home/john that start with "prefix".                           |
| <code>find /home -size +100M</code>          | # Find files larger than 100MB in /home                                        |
| <code>whereis program</code>                 | # Display the location of the binary, source and manual page files of program. |
| <code>which program</code>                   | # Display the path of executable that would run if program is executed.        |

## 12 - SSH LOGINS

|                                    |                                                                           |
|------------------------------------|---------------------------------------------------------------------------|
| <code>ssh host</code>              | # Connect to host as your local username.                                 |
| <code>ssh user@host</code>         | # Connect to host as user                                                 |
| <code>ssh -p port user@host</code> | # Connect to host using port                                              |
| <code>ssh-keygen</code>            | # Create a new SSH key pair.                                              |
| <code>ssh-copy-id user@host</code> | # Copy SSH key to the remote host to enable passwordless logins for user. |



## 13 - FILE TRANSFERS

|                                                |                                                                                               |
|------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <code>scp file.txt server:/tmp</code>          | # Secure copy file.txt to the /tmp folder on server                                           |
| <code>scp server:/var/www/*.html /tmp</code>   | # Copy *.html files from server to the local /tmp folder.                                     |
| <code>scp -r server:/var/www /tmp</code>       | # Copy all files and directories recursively from server to the current system's /tmp folder. |
| <code>rsync -a /home /backups/</code>          | # Synchronize /home to /backups/home                                                          |
| <code>rsync -avz /home server:/backups/</code> | # Synchronize files/directories between the local and remote system with compression enabled  |
| <code>ftp host</code>                          | # Connect to FTP server on the remote host.                                                   |

## 14 - DISK USAGE

|                              |                                                                             |
|------------------------------|-----------------------------------------------------------------------------|
| <code>df -h</code>           | # Show free and used space on mounted filesystems                           |
| <code>df -i</code>           | # Show free and used inodes on mounted filesystems                          |
| <code>fdisk -l</code>        | # Display disks partitions sizes and types                                  |
| <code>du -ah</code>          | # Display disk usage for all files and directories in human readable format |
| <code>du -sh</code>          | # Display total disk usage off the current directory                        |
| <code>du -a directory</code> | # Display size of all files in directory.                                   |
| <code>findmnt</code>         | # List all mounted file systems with details.                               |

## 15 - DIRECTORY NAVIGATION

|                                    |                                                                                 |
|------------------------------------|---------------------------------------------------------------------------------|
| <code>cd ..</code>                 | # To go up one level of the directory tree. (Change into the parent directory.) |
| <code>cd</code>                    | # Go to the \$HOME directory                                                    |
| <code>cd /etc</code>               | # Change to the /etc directory                                                  |
| <code>alias goto='cd /etc/'</code> | # Create goto alias for command cd /etc/.                                       |

## 16 - SECURITY

|                                 |                                                                                                                  |
|---------------------------------|------------------------------------------------------------------------------------------------------------------|
| <code>passwd</code>             | # Change the current user's password.                                                                            |
| <code>sudo -i</code>            | # Switch to the root account with root's environment. (Login shell.)                                             |
| <code>sudo -s</code>            | # Execute your current shell as root. (Non-login shell.)                                                         |
| <code>sudo -l</code>            | # List sudo privileges for the current user.                                                                     |
| <code>visudo</code>             | # Edit the sudoers configuration file.                                                                           |
| <code>getenforce</code>         | # Display the current SELinux mode.                                                                              |
| <code>sestatus</code>           | # Display SELinux details such as the current SELinux mode, the configured mode, and the loaded policy.          |
| <code>setenforce 0</code>       | # Change the current SELinux mode to Permissive. (Does not survive a reboot.)                                    |
| <code>setenforce 1</code>       | # Change the current SELinux mode to Enforcing. (Does not survive a reboot.)                                     |
| <code>SELINUX=enforcing</code>  | # Set the SELinux mode to enforcing on boot by using this setting in the <code>/etc/selinux/config</code> file.  |
| <code>SELINUX=permissive</code> | # Set the SELinux mode to permissive on boot by using this setting in the <code>/etc/selinux/config</code> file. |
| <code>SELINUX=disabled</code>   | # Set the SELinux mode to disabled on boot by using this setting in the <code>/etc/selinux/config</code> file.   |

## 17 - LOGGING AND AUDITING

|                                        |                                               |
|----------------------------------------|-----------------------------------------------|
| <code>dmesg</code>                     | # Display messages in kernel ring buffer.     |
| <code>journalctl</code>                | # Display logs stored in the systemd journal. |
| <code>journalctl -u servicename</code> | # Display logs for a specific unit (service). |